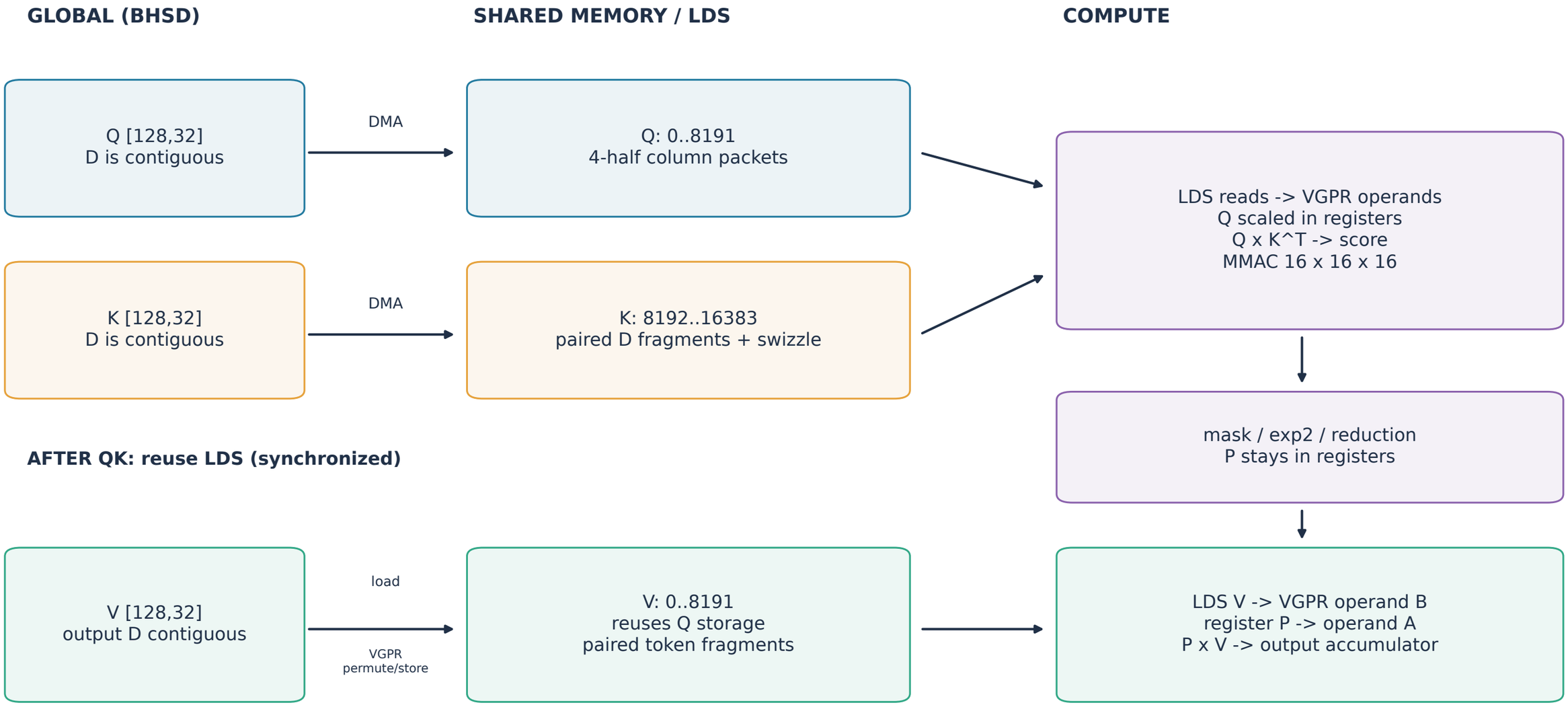
01 Global -> LDS -> lane registers -> MMAC

Read left to right. Q/K and V use the same LDS bytes at different times.



A wave cooperates on each MMAC; each lane supplies 4 half values per A/B operand.
The lane that copied a value need not be the lane that later consumes it.

Actual export: k5_hybrid_dma_guard | FP16 | 256 threads / 4 waves | Addresses in bytes | Not a timing simulation

02 Q: contiguous global load, separated LDS stores

Example: m=0, d=0..15. Same values and final LDS addresses; different copy-lane ownership.

GLOBAL: Q[m,d], byte offset = 256*m + 2*d

Q[0,0:4]
GM 0..7

Q[0,4:8]
GM 8..15

Q[0,8:12]
GM 16..23

Q[0,12:16]
GM 24..31

k5 / VGPR: copy thread 0 loads two 16-byte vectors

load Q[0,0:8] -> registers

load Q[0,8:16] -> registers

Take register slices; Q needs no cross-lane shuffle here

Q[0,0:4]
LDS 0..7
8-byte store

Q[0,4:8]
LDS 1024..1031
8-byte store

Q[0,8:12]
LDS 2048..2055
8-byte store

Q[0,12:16]
LDS 3072..3079
8-byte store

dma_guard: each 8-byte packet is written by TWO DMA lanes

DMA round 0, m0=0
threads 0, 1
4 bytes each; no data VGPR

DMA round 1, m0=1024
threads 0, 1
4 bytes each; no data VGPR

DMA round 2, m0=2048
threads 0, 1
4 bytes each; no data VGPR

DMA round 3, m0=3072
threads 0, 1
4 bytes each; no data VGPR

Q LDS address = 1024*(d//4) + 8*m + 2*(d%4) | Physical order: [d//4][m][d%4]

03 Q / K / V: what is physically adjacent in LDS?

Labels are logical coordinates, NOT lane IDs. Packet samples below include the column swizzles from the actual IR.

Q[m,d]
[4 consecutive d] per row
Q[0,0:4], Q[1,0:4], ...
next d-group starts at +1024 B

K[n,d] (used as K^T)
[4 d] + [4 d at +16]
8 half per physical packet
permuted token-column order

V[n,c]
[4 tokens] + [next 4 tokens]
8 half at one output column
permuted output-column order

K packets: labels (token n, D d)

8192	0,0	0,1	0,2	0,3	0,16	0,17	0,18	0,19
8208	1,0	1,1	1,2	1,3	1,16	1,17	1,18	1,19
8224	2,0	2,1	2,2	2,3	2,16	2,17	2,18	2,19
8240	3,0	3,1	3,2	3,3	3,16	3,17	3,18	3,19
8256	4,0	4,1	4,2	4,3	4,16	4,17	4,18	4,19
8272	5,0	5,1	5,2	5,3	5,16	5,17	5,18	5,19
8288	6,0	6,1	6,2	6,3	6,16	6,17	6,18	6,19
8304	7,0	7,1	7,2	7,3	7,16	7,17	7,18	7,19
8320	11,0	11,1	11,2	11,3	11,16	11,17	11,18	11,19
8336	8,0	8,1	8,2	8,3	8,16	8,17	8,18	8,19
8352	9,0	9,1	9,2	9,3	9,16	9,17	9,18	9,19
8368	10,0	10,1	10,2	10,3	10,16	10,17	10,18	10,19

Cell = half (2 B); row = 16 consecutive LDS bytes

V packets: labels (token n, output c)

0	0,0	1,0	2,0	3,0	4,0	5,0	6,0	7,0
16	0,1	1,1	2,1	3,1	4,1	5,1	6,1	7,1
32	0,2	1,2	2,2	3,2	4,2	5,2	6,2	7,2
48	0,3	1,3	2,3	3,3	4,3	5,3	6,3	7,3
64	0,4	1,4	2,4	3,4	4,4	5,4	6,4	7,4
80	0,5	1,5	2,5	3,5	4,5	5,5	6,5	7,5
96	0,6	1,6	2,6	3,6	4,6	5,6	6,6	7,6
112	0,7	1,7	2,7	3,7	4,7	5,7	6,7	7,7
128	0,11	1,11	2,11	3,11	4,11	5,11	6,11	7,11
144	0,8	1,8	2,8	3,8	4,8	5,8	6,8	7,8
160	0,9	1,9	2,9	3,9	4,9	5,9	6,9	7,9
176	0,10	1,10	2,10	3,10	4,10	5,10	6,10	7,10

Cell = half (2 B); row = 16 consecutive LDS bytes

Why V uses a register permutation

GLOBAL rows (contiguous D)
V[0,0:4], V[1,0:4],
V[2,0:4], V[3,0:4]

→

lane-local registers
select / permute
columns out of rows

→

LDS 8-byte stores
V[0:4,0], V[0:4,1],
V[0:4,2], V[0:4,3]

The consumer reads paired 16-byte K/V packets, then selects a 4-half operand for each MMAC.
K pairs D fragments 16 apart; V pairs token fragments 4 apart. Global order and LDS order differ.

04 Which lane supplies each MMAC operand?

Wave 0, first QK and PV instructions, traced from LLVM operands. Four half values per lane; 64 lanes together supply a 16 x 16 tile.

Q: first QK MMAC
cell number = consuming lane

query row m \ D coordinate d	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	0	0	0	16	16	16	16	32	32	32	32	48	48	48	48
1	1	1	1	1	17	17	17	17	33	33	33	33	49	49	49	49
2	2	2	2	2	18	18	18	18	34	34	34	34	50	50	50	50
3	3	3	3	3	19	19	19	19	35	35	35	35	51	51	51	51
4	4	4	4	4	20	20	20	20	36	36	36	36	52	52	52	52
5	5	5	5	5	21	21	21	21	37	37	37	37	53	53	53	53
6	6	6	6	6	22	22	22	22	38	38	38	38	54	54	54	54
7	7	7	7	7	23	23	23	23	39	39	39	39	55	55	55	55
8	8	8	8	8	24	24	24	24	40	40	40	40	56	56	56	56
9	9	9	9	9	25	25	25	25	41	41	41	41	57	57	57	57
10	10	10	10	10	26	26	26	26	42	42	42	42	58	58	58	58
11	11	11	11	11	27	27	27	27	43	43	43	43	59	59	59	59
12	12	12	12	12	28	28	28	28	44	44	44	44	60	60	60	60
13	13	13	13	13	29	29	29	29	45	45	45	45	61	61	61	61
14	14	14	14	14	30	30	30	30	46	46	46	46	62	62	62	62
15	15	15	15	15	31	31	31	31	47	47	47	47	63	63	63	63

K: first QK MMAC
cell number = consuming lane

key token n \ D coordinate d	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	0	0	0	16	16	16	16	32	32	32	32	48	48	48	48
1	4	4	4	4	20	20	20	20	36	36	36	36	52	52	52	52
2	8	8	8	8	24	24	24	24	40	40	40	40	56	56	56	56
3	12	12	12	12	28	28	28	28	44	44	44	44	60	60	60	60
16	1	1	1	1	17	17	17	17	33	33	33	33	49	49	49	49
17	5	5	5	5	21	21	21	21	37	37	37	37	53	53	53	53
18	9	9	9	9	25	25	25	25	41	41	41	41	57	57	57	57
19	13	13	13	13	29	29	29	29	45	45	45	45	61	61	61	61
32	2	2	2	2	18	18	18	18	34	34	34	34	50	50	50	50
33	6	6	6	6	22	22	22	22	38	38	38	38	54	54	54	54
34	10	10	10	10	26	26	26	26	42	42	42	42	58	58	58	58
35	14	14	14	14	30	30	30	30	46	46	46	46	62	62	62	62
48	3	3	3	3	19	19	19	19	35	35	35	35	51	51	51	51
49	7	7	7	7	23	23	23	23	39	39	39	39	55	55	55	55
50	11	11	11	11	27	27	27	27	43	43	43	43	59	59	59	59
51	15	15	15	15	31	31	31	31	47	47	47	47	63	63	63	63

V: first PV MMAC
cell number = consuming lane

key token n \ output column c	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
2	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
16	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
17	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
18	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
19	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
32	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47
33	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47
34	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47
35	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47
48	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63
49	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63
50	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63
51	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63

Follow lane 0 through two consecutive instructions

Instruction	Operand	Lane 0 logical coordinates	LDS bytes
QK #0	Q	(0,0) (0,1) (0,2) (0,3)	0..7
QK #0	K	(0,0) (0,1) (0,2) (0,3)	8192..8199
QK #1	Q	(0,16) (0,17) (0,18) (0,19)	4096..4103
QK #1	K	(0,16) (0,17) (0,18) (0,19)	8200..8207
PV #0	V	(0,0) (1,0) (2,0) (3,0)	0..7
PV #1	V	(4,0) (5,0) (6,0) (7,0)	8..15

K's selected token columns are permuted by the producer layout: read the axis coordinates, not just the grid position.
P (PV operand A) comes from registers. MMAC consumes VGPR operands; it does not read LDS directly.